

Assignment 3: Data Exploration

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Exploration.

Set up your R session

1. Load necessary packages (tidyverse, lubridate, here), check your current working directory and upload two datasets: the ECOTOX neonicotinoid dataset (ECOTOX_Neonicotinoids_Insects_raw.csv) and the Niwot Ridge NEON dataset for litter and woody debris (NEON_NIWO_Litter_massdata_2018-08_raw.csv). Name these datasets “Neonics” and “Litter”, respectively. Be sure to include the subcommand to read strings in as factors.

```
#install.packages('tidyverse')
#install.packages('lubridate')
#install.packages('here')
```

```
library(tidyverse)
library(lubridate)
library(here)
```

```
getwd()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
Neonics <- read.csv(
  file = here('Data/Raw/ECOTOX_Neonicotinoids_Insects_raw.csv'),
  stringsAsFactors = TRUE)
```

```
Litter <- read.csv(
  file = here('Data/Raw/NEON_NIWO_Litter_massdata_2018-08_raw.csv'),
  stringsAsFactors = TRUE)
```

Learn about your system

2. The neonicotinoid dataset was collected from the Environmental Protection Agency’s ECOTOX Knowledgebase, a database for ecotoxicology research. Neonicotinoids are a class of insecticides used widely

in agriculture. The dataset that has been pulled includes all studies published on insects. Why might we be interested in the ecotoxicology of neonicotinoids on insects? Feel free to do a brief internet search if you feel you need more background information.

Answer: Since neonicotinoids are one of the most commonly used insecticide in agriculture, it is important to understand their ecotoxicology to know how they will affect other organisms in the ecosystems in which they are introduced. For instance, neonicotinoids are highly toxic to bees, and their decline can trigger negative trophic cascades that can disrupt overall ecosystem balance.

3. The Niwot Ridge litter and woody debris dataset was collected from the National Ecological Observatory Network, which collectively includes 81 aquatic and terrestrial sites across 20 ecoclimatic domains. 32 of these sites sample forest litter and woody debris, and we will focus on the Niwot Ridge long-term ecological research (LTER) station in Colorado. Why might we be interested in studying litter and woody debris that falls to the ground in forests? Feel free to do a brief internet search if you feel you need more background information.

Answer: Litter and woody debris are an essential component of forest ecosystems because they play a crucial role in nutrient cycling and decomposition processes. These functions help the ecosystem to maintain a healthy state.

4. How is litter and woody debris sampled as part of the NEON network? Read the `NEON_Litterfall_UserGuide.pdf` document to learn more. List three pieces of salient information about the sampling methods here:

Answer: 1. Litter is collected in elevated 0.5m² PVC traps. 2. Woody debris is collected in ground traps. 3. Litter and woody debris are sampled at NEON sites that contain woody vegetation >2m tall.

Obtain basic summaries of your data (Neonics)

5. What are the dimensions of the dataset?

```
dim(Neonics
)
```

```
## [1] 4623  30
```

```
#The dataset has 4,623 rows and 30 columns.
```

6. Using the `summary` function on the “Effect” column, determine the most common effects that are studied. Why might these effects specifically be of interest? [Tip: The `sort()` command is useful for listing the values in order of magnitude...]

```
effect_counts <- table(Neonics$Effect)
sorted_effects <- sort(effect_counts, decreasing = TRUE)
sorted_effects
```

```
##
##      Population      Mortality      Behavior Feeding behavior
##           1803           1493           360           255
```

##	Reproduction	Development	Avoidance	Genetics
##	197	136	102	82
##	Enzyme(s)	Growth	Morphology	Immunological
##	62	38	22	16
##	Accumulation	Intoxication	Biochemistry	Cell(s)
##	12	12	11	9
##	Physiology	Histology	Hormone(s)	
##	7	5	1	

```
head(sorted_effects, 3)
```

```
##
## Population Mortality Behavior
##      1803      1493      360
```

Answer: The three most common effects that are studied are population, mortality, and behavior. These are critical elements of population dynamics, and can be indicative of either positive or negative effects of insecticides on the population of an ecosystem.

- Using the `summary` function, determine the six most commonly studied species in the dataset (common name). What do these species have in common, and why might they be of interest over other insects? Feel free to do a brief internet search for more information if needed. [TIP: Explore the help on the `summary()` function, in particular the `maxsum` argument...]

```
species.common.name_counts <- table(Neonics$Species.Common.Name)
sorted_species.common.name <- sort(species.common.name_counts, decreasing = TRUE)
sorted_species.common.name
```

```
##
##              Honey Bee              Parasitic Wasp
##              667              285
##      Buff Tailed Bumblebee      Carniolan Honey Bee
##              183              152
##              Bumble Bee              Italian Honeybee
##              140              113
##      Japanese Beetle              Asian Lady Beetle
##              94              76
##      Euonymus Scale              Wireworm
##              75              69
##      European Dark Bee              Minute Pirate Bug
##              66              62
##      Asian Citrus Psyllid              Parastic Wasp
##              60              58
##      Colorado Potato Beetle              Parasitoid Wasp
##              57              51
##      Erythrina Gall Wasp              Beetle Order
##              49              47
##      Snout Beetle Family, Weevil              Sevenspotted Lady Beetle
##              47              46
##      True Bug Order              Buff-tailed Bumblebee
##              45              39
##      Aphid Family              Cabbage Looper
```

##	38	38
##	Sweetpotato Whitefly	Braconid Wasp
##	37	33
##	Cotton Aphid	Predatory Mite
##	33	33
##	Ladybird Beetle Family	Parasitoid
##	30	30
##	Scarab Beetle	Spring Tiphia
##	29	29
##	Thrip Order	Ground Beetle Family
##	29	27
##	Rove Beetle Family	Tobacco Aphid
##	27	27
##	Chalcid Wasp	Convergent Lady Beetle
##	25	25
##	Stingless Bee	Spider/Mite Class
##	25	24
##	Tobacco Flea Beetle	Citrus Leafminer
##	24	23
##	Ladybird Beetle	Mason Bee
##	23	22
##	Mosquito	Argentine Ant
##	22	21
##	Beetle	Flatheaded Appletree Borer
##	21	20
##	Horned Oak Gall Wasp	Leaf Beetle Family
##	20	20
##	Potato Leafhopper	Tooth-necked Fungus Beetle
##	20	20
##	Codling Moth	Black-spotted Lady Beetle
##	19	18
##	Calico Scale	Fairyfly Parasitoid
##	18	18
##	Lady Beetle	Minute Parasitic Wasps
##	18	18
##	Mirid Bug	Mulberry Pyralid
##	18	18
##	Silkworm	Vedalia Beetle
##	18	18
##	Araneoid Spider Order	Bee Order
##	17	17
##	Egg Parasitoid	Insect Class
##	17	17
##	Moth And Butterfly Order	Oystershell Scale Parasitoid
##	17	17
##	Hemlock Woolly Adelgid Lady Beetle	Hemlock Woolly Adelgid
##	16	16
##	Mite	Onion Thrip
##	16	16
##	Western Flower Thrips	Corn Earworm
##	15	14
##	Green Peach Aphid	House Fly
##	14	14
##	Ox Beetle	Red Scale Parasite

##	14	14
##	Spined Soldier Bug	Armoured Scale Family
##	14	13
##	Diamondback Moth	Eulophid Wasp
##	13	13
##	Monarch Butterfly	Predatory Bug
##	13	13
##	Yellow Fever Mosquito	Braconid Parasitoid
##	13	12
##	Common Thrip	Eastern Subterranean Termite
##	12	12
##	Jassid	Mite Order
##	12	12
##	Pea Aphid	Pond Wolf Spider
##	12	12
##	Spotless Ladybird Beetle	Glasshouse Potato Wasp
##	11	10
##	Lacewing	Southern House Mosquito
##	10	10
##	Two Spotted Lady Beetle	Ant Family
##	10	9
##	Apple Maggot	Asiatic Honey Bee
##	9	9
##	Eulophid Parasitoid	Lacewing Family
##	9	9
##	Mealybug Destroyer	Alfalfa Leafcutter Bee
##	9	8
##	Bee	Bumblebee
##	8	8
##	Chilean Predatory Mite	Dwarf Honey Bee
##	8	8
##	Neotropical Stingless Bee	Parasitic Wasp Family
##	8	8
##	Spiralling Whitefly	Beetle Mite Family
##	8	7
##	Chinch Bug	Macedonian Honey Bee
##	7	7
##	Moth	Potato Tuberworm
##	7	7
##	Russian Wheat Aphid	Soldier Beetle
##	7	7
##	Southern One-Year Canegrub	Tarnished Plant Bug
##	7	7
##	Ambrosia Beetle	Aphid Wasp
##	6	6
##	Black Vine Weevil	Childers Canegrub
##	6	6
##	Coconut Leaf Beetle	Eleven-spotted Ladybird Beetle
##	6	6
##	Encyrtid Wasp	European Red Mite
##	6	6
##	Fall Armyworm	Fruit Fly
##	6	6
##	Hover Fly	Oblique Banded Leaf Roller

##	6	6
##	Obscure Mealybug	Oribatid Mite Suborder
##	6	6
##	Pistachio Psyllid	Redbay Ambrosia Beetle
##	6	6
##	Silverleaf Whitefly	Soybean Aphid
##	6	6
##	Subterranean Termite	Thrip
##	6	6
##	Two-Spotted Spider Mite	Apple Aphid
##	6	5
##	Brown Planthopper	Earwig
##	5	5
##	Green June Beetle	Hornfaced Bee
##	5	5
##	Long Horned Beetle Family	Plum Curculio
##	5	5
##	Rove Beetle	San Jose Scale
##	5	5
##	Scelionid Wasp	Speckled Cutworm Moth
##	5	5
##	Thrip Family	Ant
##	5	4
##	Cabbage Seedpod Weevil	Common Green Lacewing
##	4	4
##	Eucalyptus Gall Wasp	European Apple Sawfly
##	4	4
##	European Honey Bee	European Tarnished Plant Bug
##	4	4
##	Garden Symphytan	Linyphiid Spider
##	4	4
##	Onion Maggot	Oriental Beetle
##	4	4
##	Parsnip Seed Wasp	Pea And Bean Weevil
##	4	4
##	Pear Sucker	Red Imported Fire Ant
##	4	4
##	Striped Cucumber Beetle	Sugarcane Beetle
##	4	4
##	Wasp	Wolf Spider Family
##	4	4
##	Yellow-faced Bumblebee	Ambrosia Bark Beetle
##	4	3
##	Asian Ambrosia Beetle	Beetle Family
##	3	3
##	Birch Leafminer	Black Twig Borer
##	3	3
##	Braconid Parasitoid Wasp	California Red Scale
##	3	3
##	Crucifer Flea Beetle	Cutworm
##	3	3
##	Delphacid Planthopper	Egyptian Cotton Leafworm
##	3	3
##	Encyrtid Parasitoid	Fly/Mosquito/Midge Order

##		3		3
##	Formosan Subterranean Termite		Fruit-tree Pinhole Borer	
##		3		3
##	Green Rice Leafhopper		Ground Beetle	
##		3		3
##	Ichneumonid Wasp		Large-Jawed Orb Weaver Family	
##		3		3
##	Leaf Cutting Ant		Mediterranean Fruit Fly	
##		3		3
##	Minute Flour Bug		Mite Family	
##		3		3
##	Moth Family		Negatoria Canegrub	
##		3		3
##	Sap Beetle Family		Scale Insect Order	
##		3		3
##	Scarab Beetle Family		Sheet-Web Weaver Family	
##		3		3
##	Spider		Sugarcane Grub	
##		3		3
##	Tenebrionid Beetle		Alfalfa Plant Bug	
##		3		2
##	Alkali Bee		Aphid	
##		2		2
##	Assassin Bug		Azalea Lace Bug	
##		2		2
##	Banana Aphid		Brown Scale	
##		2		2
##	Brown Stinkbug		Budworm	
##		2		2
##	Cabbage Aphid		Cabbage White	
##		2		2
##	Cardamom Thrip		Carrot Weevil	
##		2		2
##	Celer Crab Spider		Centipede Class	
##		2		2
##	Citricola Scale		Clouded Plant Bug	
##		2		2
##	Coffee Bean Weevil		Cotton Fleahopper	
##		2		2
##	Egyptian Alfalfa Weevil		Engraver Beetle	
##		2		2
##	Fig Longicorn Beetle		Glassy-winged Sharpshooter	
##		2		2
##	Hawthorn Lace Bug		Hister Beetle Family	
##		2		2
##	Jumping Spider Family		Lined Click Beetle	
##		2		2
##	Maple Spider Mite		Meshweaver Spider	
##		2		2
##	Minute Pirate Bug Family		Predaceous Fly	
##		2		2
##	Pygmy Mangold Beetle		Rose Sawfly	
##		2		2
##	Serpentine Leafminer		Spider Mite Destroyer	

##		2		2
##	Spotted Tentiform Leafminer			Stink Bug
##		2		2
##	Tawny Mole Cricket		Tick/Chigger/Mite Order	
##		2		2
##	Turf Running-spider		Turnip Aphid	
##		2		2
##	Western Bigeyed Bug		Western Damsel Bug	
##		2		2
##	Western Plant Bug		White Apple Leafhopper Nymph	
##		2		2
##	White-backed Planthopper		Whitemarked Fleahopper	
##		2		2
##	Antlike Flower Beetle		Banded Soft-winged Flower Beetle	
##		1		1
##	Banded Sunflower Moth		Bee Family	
##		1		1
##	Beet Armyworm		Black Citrus Aphid	
##		1		1
##	Blue Alfalfa Aphid		Cabbage Root Fly	
##		1		1
##	Cactus Lady Beetle		Citrus Red Mite	
##		1		1
##	Cottony Cushion Sale		Crapemyrtle Aphid	
##		1		1
##	Damselbug Family		Ectoparasitoid Wasp	
##		1		1
##	English Grain Aphid		Fairyfly	
##		1		1
##	Flea Beetle		Gall Midge	
##		1		1
##	Grasshopper/Cricket/Locust Order		Greenhouse Whitefly	
##		1		1
##	Grey Sunflower Seed Weevil		Harvestman Spider Order	
##		1		1
##	Hawthorn Leaf Miner		Longtailed Fruit Fly Parasite	
##		1		1
##	Minute Lady Beetles		Painted Maple Aphid	
##		1		1
##	Pepper Weevil		Pine False Webworm	
##		1		1
##	Plant Bug		Pollen Beetle	
##		1		1
##	Predacious Mite		Predator Bug	
##		1		1
##	Pseudocentipede Class		Pteromalid Wasp Family	
##		1		1
##	Red Sunflower Seed Weevil		Rice Leaf Folder Moth	
##		1		1
##	Rose Grain Aphid		Scale Picnic Beetle	
##		1		1
##	Shiny Spider Beetle		Southern Army Worm	
##		1		1
##	Spirea Aphid		Spotted Sunflower Stem Weevil	


```
##              1              1
## Strawberry Blossom Weevil      Sunflower Midge
##              1              1
##              Sunflower Moth      Ten-spot Ladybird Beetle
##              1              1
##              Tobacco Thrip      Twicestabbed Lady Beetle
##              1              1
##              Wasp Family          Weevil
##              1              1
## Yellow Mealworm Beetle
##              1
```

Answer: The six most commonly studied species in the dataset are the honey bee, parasitic wasp, buff tailed bumblebee, carniolan honey bee, bumble bee, and italian honeybee. These are all bee species; these are likely of interest over other species because of the negative effect of neonicotinoids on bees.

8. Concentrations are always a numeric value. What is the class of `Conc.1..Author.` column in the dataset, and why is it not numeric? [Tip: Viewing the dataframe may be helpful...]

```
class(Neonics$Conc.1..Author.)
```

```
## [1] "factor"
```

Answer: The `Conc.1..Author` column is a factor class. This is because some of the values in this column contain less than or greater than symbols, which is a non-numeric character; this makes the column a factor instead of a numeric class.

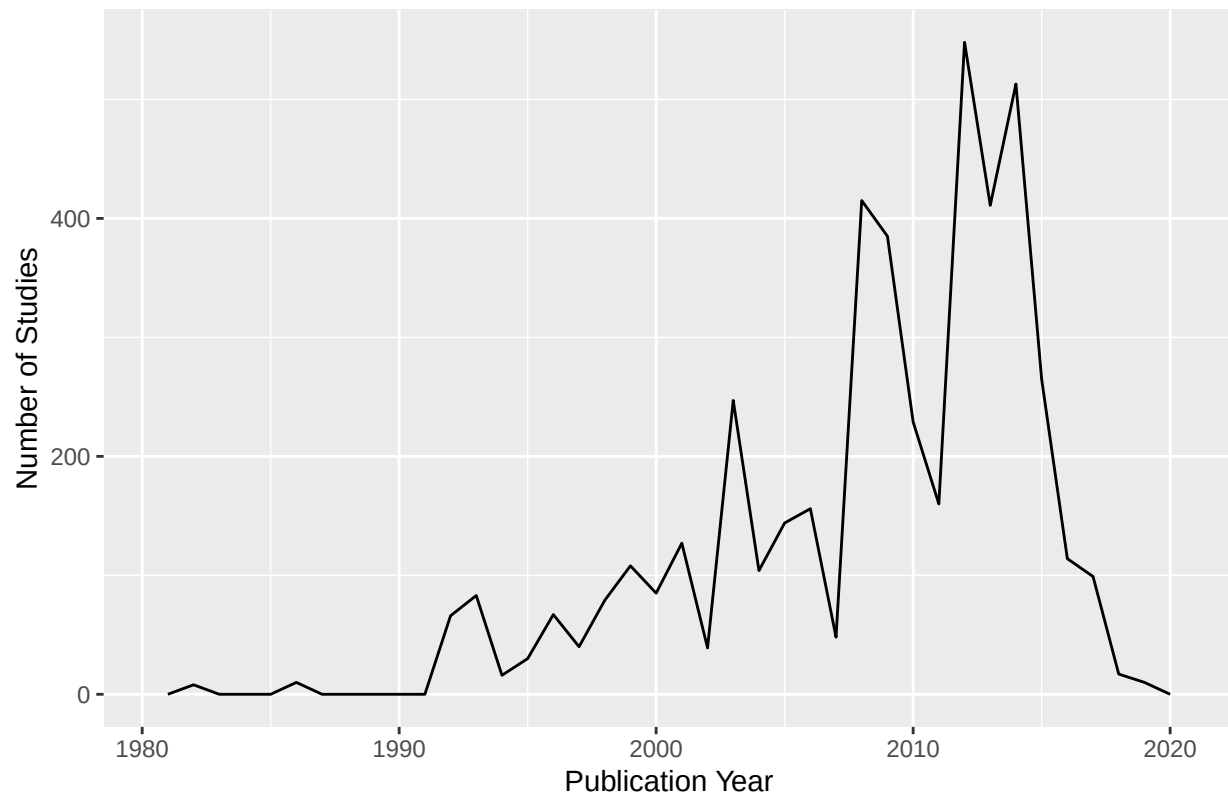
Explore your data graphically (Neonics)

9. Using `geom_freqpoly`, generate a plot of the number of studies conducted by publication year.

```
library(ggplot2)

ggplot(Neonics, aes(x = Publication.Year)) +
  geom_freqpoly(binwidth = 1) +
  labs(
    title = "Number of Studies by Publication Year",
    x = "Publication Year",
    y = "Number of Studies"
  )
```

Number of Studies by Publication Year

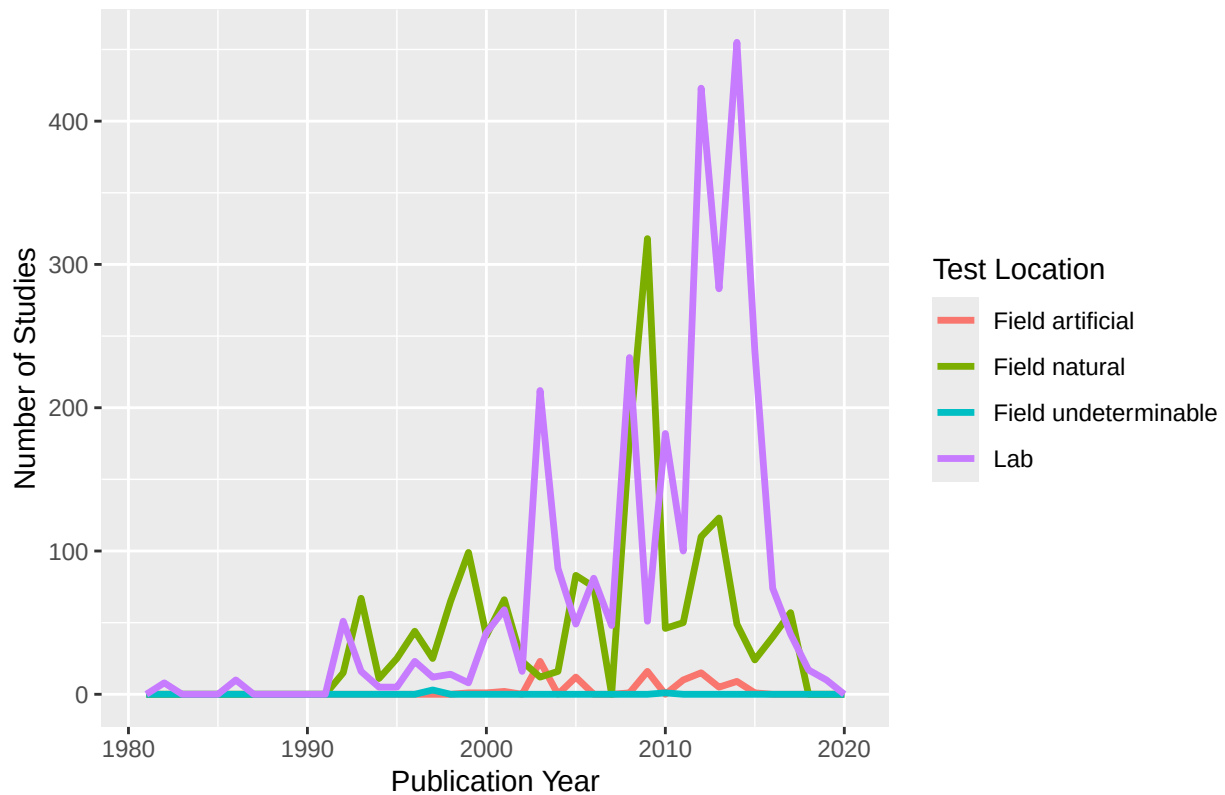


10. Reproduce the same graph but now add a color aesthetic so that different Test.Location are displayed as different colors.

```
ggplot(Neonics, aes(x = Publication.Year, color = Test.Location)) +
  geom_freqpoly(binwidth = 1, size = 1.2) +
  labs(
    title = "Number of Studies by Publication Year and Test Location",
    x = "Publication Year",
    y = "Number of Studies",
    color = "Test Location"
  )
```

```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

Number of Studies by Publication Year and Test Location



Interpret this graph. What are the most common test locations, and do they differ over time?

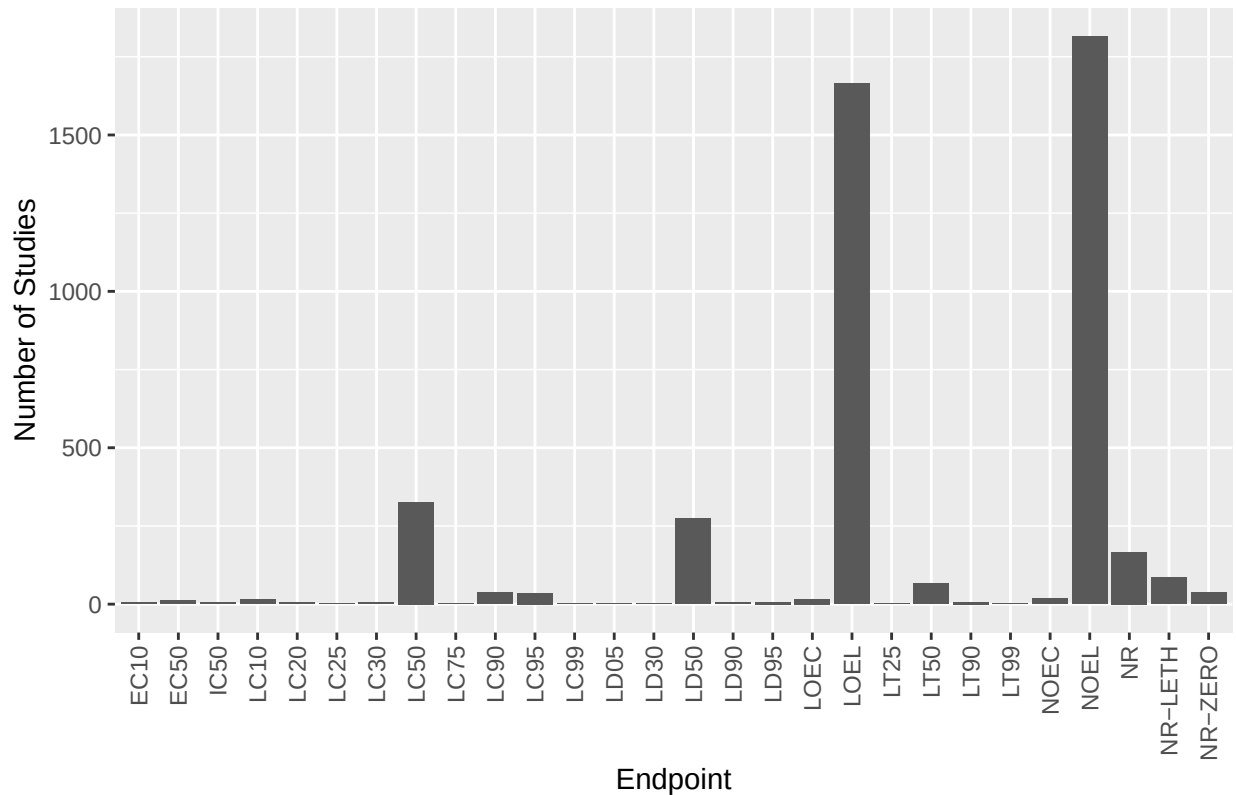
Answer: The most common test locations are the lab, undeterminable field, natural field and artificial field. The natural field location was most common around 2010. Over time, the lab steadily increased as the most common test location.

11. Create a bar graph of Endpoint counts. What are the two most common end points, and how are they defined? Consult the ECOTOX_CodeAppendix for more information.

[**TIP:** Add `theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))` to the end of your plot command to rotate and align the X-axis labels...]

```
ggplot(Neonics, aes(x = Endpoint)) +
  geom_bar() +
  labs(
    title = "Counts of Endpoints in Neonics Studies",
    x = "Endpoint",
    y = "Number of Studies"
  ) +
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust = 1))
```

Counts of Endpoints in Neonic Studies



Answer: The two most common endpoints are LOEL and NOEL. According to the Ecotox Code Appendix, LOEL stands for “Lowest-observable-effect-level” and is used for terrestrial areas. NOEL stands for “No-observable-effect-level” and is used for terrestrial areas.

Explore your data (Litter)

- Determine the class of collectDate. Is it a date? If not, change to a date and confirm the new class of the variable. Using the `unique` function, determine which dates litter was sampled in August 2018.

```
class(Litter$collectDate)

## [1] "factor"

#This is currently a factor class.

Litter$collectDate <- as.Date(Litter$collectDate, format = "%Y-%m-%d")
august2018_dates <- unique(Litter$collectDate[format(Litter$collectDate, "%Y-%m") == "2018-08"])
august2018_dates

## [1] "2018-08-02" "2018-08-30"
```

#The sampling dates of August 2018 are 8/2/2018 and 8/30/2018.

13. Using the `unique` function, list the different plotIDs sampled at Niwot Ridge. How is the information obtained from `unique` different from that obtained from `summary`?

```
unique_plotIDs <- unique(Litter$plotID[Litter$siteID == "NIWO"])
unique_plotIDs
```

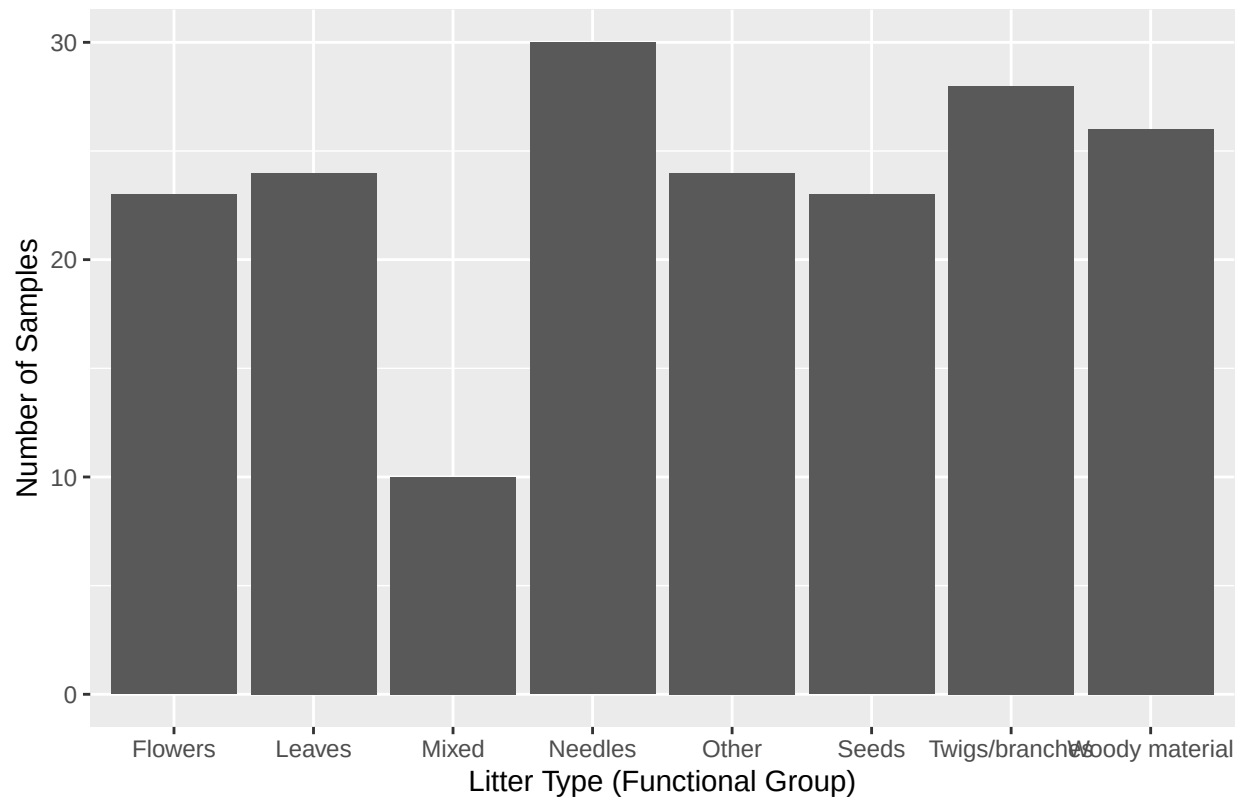
```
## [1] NIWO_061 NIWO_064 NIWO_067 NIWO_040 NIWO_041 NIWO_063 NIWO_047 NIWO_051
## [9] NIWO_058 NIWO_046 NIWO_062 NIWO_057
## 12 Levels: NIWO_040 NIWO_041 NIWO_046 NIWO_047 NIWO_051 NIWO_057 ... NIWO_067
```

Answer: The information obtained from the ‘unique’ function gives you all of the exact values within the dataset that you are searching for. The ‘summary’ function would give you a summary of the values instead of distinct values.

14. Create a bar graph of functionalGroup counts. This shows you what type of litter is collected at the Niwot Ridge sites. Notice that litter types are fairly equally distributed across the Niwot Ridge sites.

```
niwot_data <- Litter[Litter$siteID == "NIWO", ]
ggplot(niwot_data, aes(x = functionalGroup)) +
  geom_bar() +
  labs(
    title = "Counts of Functional Groups at Niwot Ridge",
    x = "Litter Type (Functional Group)",
    y = "Number of Samples"
  )
```

Counts of Functional Groups at Niwot Ridge

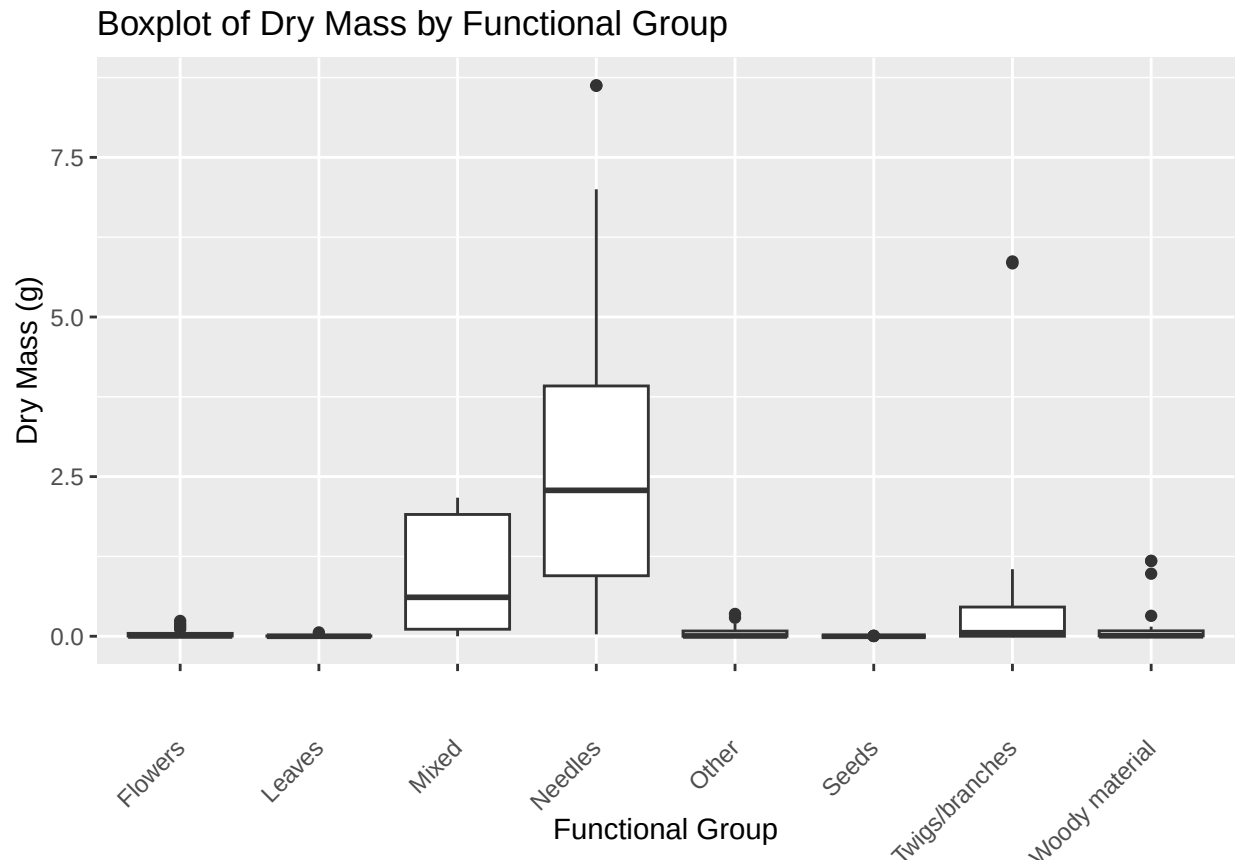


```
theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust = 1))
```

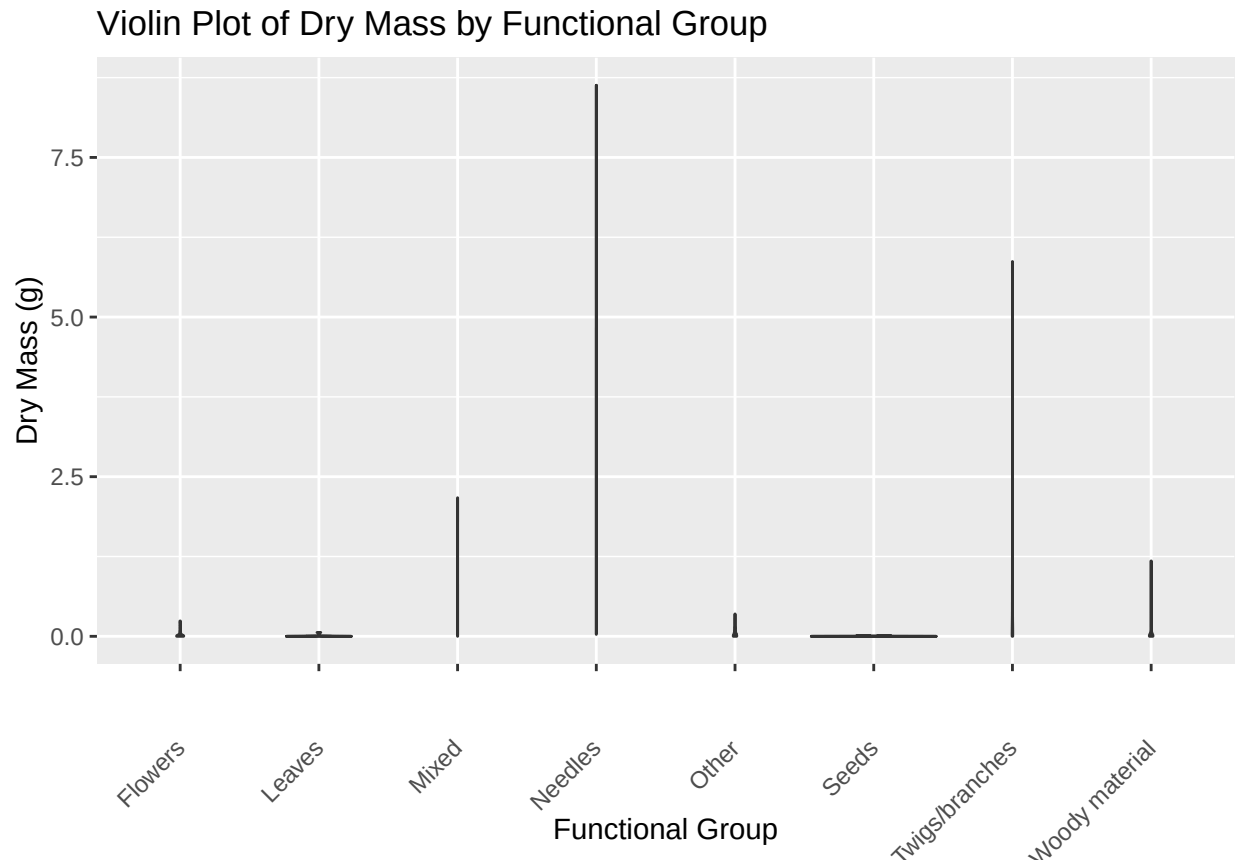
```
## List of 1
## $ axis.text.x:List of 11
## ..$ family      : NULL
## ..$ face        : NULL
## ..$ colour      : NULL
## ..$ size        : NULL
## ..$ hjust       : num 1
## ..$ vjust       : num 0.5
## ..$ angle       : num 90
## ..$ lineheight  : NULL
## ..$ margin      : NULL
## ..$ debug       : NULL
## ..$ inherit.blank: logi FALSE
## ..- attr(*, "class")= chr [1:2] "element_text" "element"
## - attr(*, "class")= chr [1:2] "theme" "gg"
## - attr(*, "complete")= logi FALSE
## - attr(*, "validate")= logi TRUE
```

- Using `geom_boxplot` and `geom_violin`, create a boxplot and a violin plot of `dryMass` by functional-Group.

```
ggplot(Litter, aes(x = functionalGroup, y = dryMass)) +
  geom_boxplot() +
  labs(
    title = "Boxplot of Dry Mass by Functional Group",
    x = "Functional Group",
    y = "Dry Mass (g)"
  ) +
  theme(axis.text.x = element_text(angle = 45, vjust = 0.5, hjust = 1))
```



```
ggplot(Litter, aes(x = functionalGroup, y = dryMass)) +
  geom_violin() +
  labs(
    title = "Violin Plot of Dry Mass by Functional Group",
    x = "Functional Group",
    y = "Dry Mass (g)"
  ) +
  theme(axis.text.x = element_text(angle = 45, vjust = 0.5, hjust = 1))
```



Why is the boxplot a more effective visualization option than the violin plot in this case?

Answer: The boxplot gives a more clear visualization of the dry mass values of the functional groups. It also includes information on the median, quartiles, and outliers, which is helpful when interpreting the data.

What type(s) of litter tend to have the highest biomass at these sites?

Answer: Needles, twigs/branches, and mixed groups of litter tend to have the highest biomasses.